Time Dilation

Measurements of time intervals are affected by the relative motion between an observer and what is observed. As a result, a clock that moves with respect to an observer ticks more slowly than it does without such motion, and all processes occur more slowly to an observer when they take place in a different inertial frames.

If someone in a moving spacecraft finds that the time interval between two events in the spacecraft is t_0, we on the ground would find that the same interval has the longer duration t. The quantity t_0, which is determined by events that occur *at the same* place in an observer's frame of reference, is called the *proper time* of the interval between the events. When witnessed from the ground, the events that mark the beginning and end of the time interval occur at different places, and in consequence the duration of the interval appears longer than the proper time. This effect is called *time dilation*.

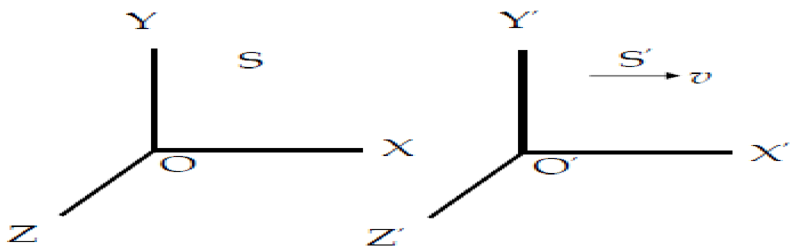

Fig. 1.

Let us consider two inertial systems S and $S^{/}$. $S^{/}$ moves to the right along x-axis with respect to S with a uniform velocity v. Again the observers O and $O^{/}$ in the inertial systems S and $S^{/}$ measures the time interval between any two events by their own clocks.

The observer O at a fixed point x_1 in the frame S measures the time interval to that elapses between two events (two signals) that occur at times t_1 and t_2.

$$t_0 = t_2 - t_1 \tag{1}$$

Where t_0 is called proper time.

Let the time registered by the observer $O^{/}$ in $S^{/}$ are $t_1^{/}$ and $t_2^{/}$.Thus moving observer $O^{/}$ in the inertial system $S^{/}$ measures the time $t_2^{/} - t_1^{/} = t$ between the same events for which the observer O in the system measures the time interval,

$$t = t_2^{/} - t_1^{/} \tag{2}$$

According Lorentz transformations

$$t_1^{/} = \frac{t_1 - \frac{v}{c^2}x_1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$t_2^{/} = \frac{t_2 - \frac{v}{c^2}x_2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

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The clock in the system S remains fixed at the point x_1.

$$x_2 = x_1$$

$$\text{and} \qquad t_2^{/} - t_1^{/} = \frac{t_2 - t_1}{\sqrt{1 - \frac{v^2}{c^2}}} \qquad (3)$$

Combining the above equations, we have

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

So the interval of time observed in a moving frame of reference will be less than the same interval of time observed in a stationary frame of reference. The phenomenon is known as *time dilation*.

Relativity of mass

Relativistic mass: It is the mass of a body in motion relative to the observer. It is equal to the rest mass multiplied by a factor that is greater than 1 and that increases as the magnitude of the velocity increases.

The relativistic mass is a function of the rest mass and the velocity of the object. If the rest mass of a body is m_0 then the mass of the body in motion is

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Variation of mass of a particle with velocity: Let us consider an elastic collision between two particles A and B located in frames of reference S and $S^{/}$ respectively. The frame $S^{/}$ is moving in the positive x-axis with respect to S at the velocity V.

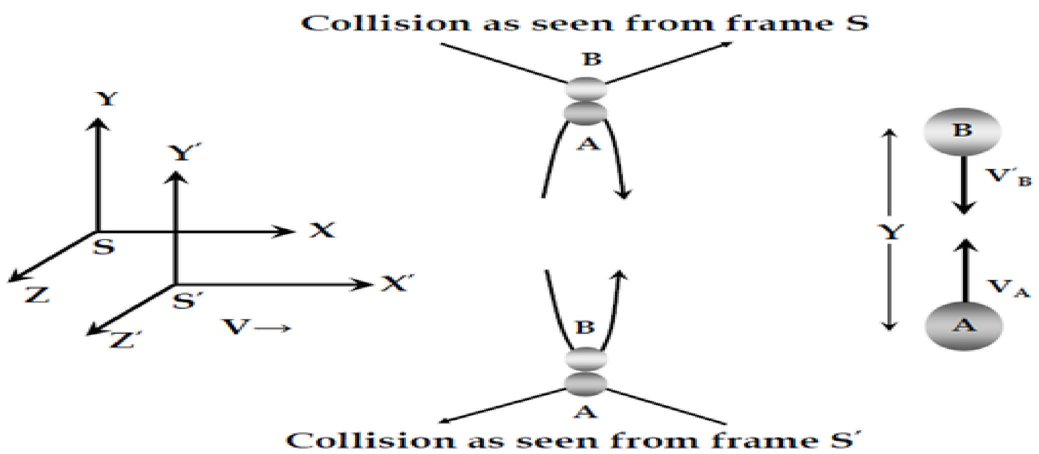

Fig. 2.

At any instant, A was thrown in the $+y$-direction at the speed V_A while B was thrown in the $-y^{/}$-direction at the speed $V^{/}_B$, where,

$$V_A = V_B^{/} \qquad (1)$$

Hence the behaviour of A as seen from S exactly the same as the behaviour of B seen from $S^{/}$.

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When the two particles collide, A rebounds in the $-y^{/}$-direction at the speed V_A while B rebounds in the $+y^{/}$-direction at speed $V^{/}_B$. If the particles are thrown from position Y apart, an observer in S finds that the collision occurs at $y = \frac{1}{2}Y$ and one in S$^{/}$ finds that the collision occurs at $y = y^{/} = \frac{1}{2}Y$. Thus, the round-trip time T_0 for A as measured in frame S,

$$T_0 = \frac{Y}{V_A} \tag{2}$$

and it is the same for B as measured in frame $S^{/}$,

$$T_0 = \frac{Y}{V_B^{/}} \tag{3}$$

If linear momentum is conserved in the S frame then from conservation of principle of linear momentum, we get

$$m_A V_A = m_B V_B \tag{4}$$

where m_A and m_B are the masses of A and B, and V_A and V_B are their velocities in the S frame.

Now if the round-trip time is T for B in S-frame, then

$$V_B = \frac{Y}{T} \tag{5}$$

In S$^{/}$, B's trip requires the time T_0. Where

$$T = \frac{T_0}{\sqrt{1-\frac{v^2}{c^2}}} \tag{6}$$

Substituting this value in equation (5) we get

$$V_B = \frac{Y\sqrt{1-\frac{v^2}{c^2}}}{T_0} \tag{7}$$

Now putting the values of V_A and V_B in eqn (4), we get,

$$m_A \frac{Y}{T_0} = m_B \frac{Y\sqrt{1-\frac{v^2}{c^2}}}{T_0}$$

$$m_A = m_B \sqrt{1-\frac{v^2}{c^2}}$$

If we put $m_A = m_0 \; and \; m_B = m_0$, then we can write

$$m_0 = m\sqrt{1-\frac{v^2}{c^2}}$$

$$m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} \tag{8}$$

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When a particle moves with the speed of light, then $v = c$. So that we can write

$$m = \frac{m_0}{\sqrt{1 - \frac{c^2}{c^2}}} = \frac{m_0}{\sqrt{1-1}} = \frac{m_0}{0}$$

$$\therefore \qquad m = \infty$$

Thus *the mass of a particle becomes infinite when it moves with the speed of light*.

Mass-Energy relation

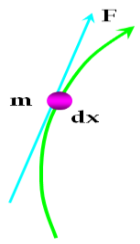
If a body of mass *m* moving with a velocity *v* has a force applied to it, then from Newton's 2nd law of motion, we get

$$F = \frac{d}{dt}(mv)$$

According to the theory of relativity, mass and velocity both are variables. Therefore,

$$F = \frac{d}{dt}(mv) = m\frac{dv}{dt} + v\frac{dm}{dt} \qquad (1)$$

Let the force *F* displace the body through a distance *dx*. Then the increase in the kinetic energy (dE_k) of the body is equal to the work done (Fdx). Hence,



$$dE_k = Fdx = m\frac{dv}{dt}dx + v\frac{dm}{dt}dx$$

$$dE_k = m\frac{dx}{dt}dv + v\frac{dx}{dt}dm$$

$$dE_k = mvdv + v^2dm \qquad (2)$$

According to the law of variation of mass with velocity

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m^2 = \frac{m_0^2}{1 - \frac{v^2}{c^2}}$$

$$m^2 = \frac{m_0^2c^2}{c^2 - v^2}$$

$$m^2c^2 - m^2v^2 = m_0^2c^2$$

$$\therefore \qquad m^2c^2 = m_0^2c^2 + m^2v^2 \qquad (3)$$

Differentiating both sides of the above equation we get

$$c^2 2m\,dm = v^2 2m\,dm + m^2 2v\,dv$$

$$\therefore \qquad c^2dm = v^2dm + mv\,dv \qquad (4)$$

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Using equation (2), equation (4) becomes

$$dE_k = c^2 dm$$

Now consider the mass of a body at rest is m_0 and when it is moving with a velocity v it is m. Integrating both sides, we get,

$$\int_0^{E_k} dE_k = c^2 \int_{m_0}^{m} dm$$

$$E_k = c^2(m - m_0)$$

$$\therefore \qquad E_k = mc^2 - m_0c^2 \qquad (5)$$

This is the relativistic formula for kinetic energy.

When the body is at rest, the energy stored in the body is m_0c^2, which the rest mass energy. The total energy of a body is the sum of the kinetic energy and rest mass energy.

$$Total\ energy = Kinetic\ energy + \ rest\ mass\ energy$$

$$E = (mc^2 - m_0c^2) + m_0c^2$$

$$\therefore \qquad E = mc^2 \qquad (6)$$

Equation (6) is the Einstein's mass energy relation.

<u>Energy-Momentum relation</u>

Let us consider a body of rest mass m_0 is moving with a velocity v. The energy and momentum of the body is given by

$$E = mc^2$$

$$P = mv$$

We know, the relativistic formula for the variation of mass with velocity is

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Hence, we can write

$$E = \frac{m_0c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \qquad (1)$$

$$P = \frac{m_0v}{\sqrt{1 - \frac{v^2}{c^2}}} \qquad (2)$$

Multiplying equation (2) by c, we get

$$Pc = \frac{m_0vc}{\sqrt{1 - \frac{v^2}{c^2}}} \qquad (3)$$

Squaring equations (1) and (3) we get

$$E^2 = \frac{m_0^2c^4}{1 - \frac{v^2}{c^2}} \qquad (4)$$

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$$P^2c^2 = \frac{m_0^2 v^2 c^2}{1 - \frac{v^2}{c^2}} \qquad (5)$$

Subtracting equation (5) from (4) we get

$$E^2 - P^2c^2 = \frac{m_0^2 c^4}{1 - \frac{v^2}{c^2}} - \frac{m_0^2 v^2 c^2}{1 - \frac{v^2}{c^2}}$$

$$E^2 - P^2c^2 = \frac{m_0^2 c^4}{1 - \frac{v^2}{c^2}}\left(1 - \frac{v^2}{c^2}\right)$$

$$E^2 - P^2c^2 = m_0^2 c^4$$

$$\therefore \qquad E^2 = P^2c^2 + m_0^2 c^4 \qquad (6)$$

Equation (6) is the energy-momentum relation.

For a massless particle, $m_0 = 0$. So that we can write

$$E = Pc \qquad (7)$$

This is the expression for the energy of a massless particle.